ELEVATE LABS — INTERNSHIP PROJECT REPORT

Project 1: Customer Lifetime Value (CLV) Prediction Model

1. Introduction

Customer Lifetime Value (CLV) is the estimated value a customer may generate for a business over a future period. Predicting CLV helps organisations understand customer value, plan retention activities and allocate marketing resources effectively.

2. Abstract

This project develops a machine-learning workflow to predict each customer's future 12-month value using historical purchase behaviour. Features include purchase frequency, recency, average order value, customer tenure, historical monetary value, discount rate and returns rate. Regression models are trained and compared, after which customers are grouped into Low, Medium and High Value segments. The dataset supplied with this project is synthetic and is used for demonstration.

3. Tools Used

Python: Pandas and NumPy for data handling; Scikit-learn for model building and evaluation; Matplotlib for visualisation. **Models:** Random Forest Regressor and Histogram-based Gradient Boosting Regressor. **Deliverables:** prediction CSV, model metrics and charts.

4. Steps Involved in Building the Project

• **Data preparation:** Load the customer dataset and inspect data types, missing values and duplicate records.

• **Feature selection:** Select frequency, recency, average order value, tenure, historical spend, discount rate and returns rate as input features.

• **Model development:** Split the data into training and testing sets; train Random Forest and Gradient Boosting regression models.

• **Model evaluation:** Compare performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R-squared (R^2).

• **Customer segmentation:** Predict 12-month CLV and group customers into Low, Medium and High Value categories.

• **Visualisation:** Create actual-versus-predicted and customer-segment charts.

5. Conclusion

The project demonstrates how predictive analytics can support customer-focused decisions. CLV predictions can help prioritise loyalty programmes, retention campaigns and cross-selling activities. For real business use, the model must be validated using authorised transaction data and monitored over time.

Project outputs: customer dataset, Python training script, model comparison metrics, predicted CLV CSV and visualisations. **Data limitation:** the included sample records are generated for demonstration and are not real company data.

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Project 2: HR Analytics — Predict Employee Attrition

1. Introduction

Employee attrition occurs when employees leave an organisation. High attrition can increase recruitment and training costs and affect team continuity. HR analytics helps organisations study workforce patterns and identify areas where employee retention efforts may be needed.

2. Abstract

This project analyses employee information and builds a classification model to estimate the likelihood of employee attrition. The dataset includes age, department, monthly income, distance from home, years at the company, overtime, job satisfaction, years since last promotion and work-life balance. The workflow covers data exploration, preprocessing, model training, evaluation and risk-band generation. The dataset is synthetic and intended for learning.

3. Tools Used

Python: Pandas for data handling; Scikit-learn for preprocessing, classification and evaluation; Matplotlib for charts. **Model:** class-balanced Random Forest Classifier. **Evaluation:** accuracy, precision, recall, F1-score and confusion matrix. Permutation feature importance is used to interpret predictive performance.

4. Steps Involved in Building the Project

• **Data preparation:** Load the HR dataset and inspect missing values, duplicates, data types and the attrition distribution.

• **Exploratory analysis:** Examine attrition patterns by department, overtime, satisfaction, income and other employee factors.

• **Preprocessing:** Impute missing values, scale numerical variables and one-hot encode categorical variables.

• **Model development:** Split the dataset into training and testing sets and train a class-balanced Random Forest classifier.

• **Model evaluation:** Evaluate using accuracy, precision, recall, F1-score and a confusion matrix.

• **Interpretation:** Calculate permutation feature importance and generate Low, Medium and High predicted-risk bands for review.

5. Conclusion

The project demonstrates how HR analytics can reveal workforce patterns and inform retention initiatives. Findings may guide supportive discussions about workload, job satisfaction, career development and work-life balance. Predictions should support—not replace—professional HR judgement and must never be the sole basis for employment decisions.

Project outputs: employee dataset, Python training script, model metrics, classification report, confusion matrix, feature-importance chart and employee risk-score CSV. **Data limitation:** the sample records are synthetic and do not represent actual employees.